

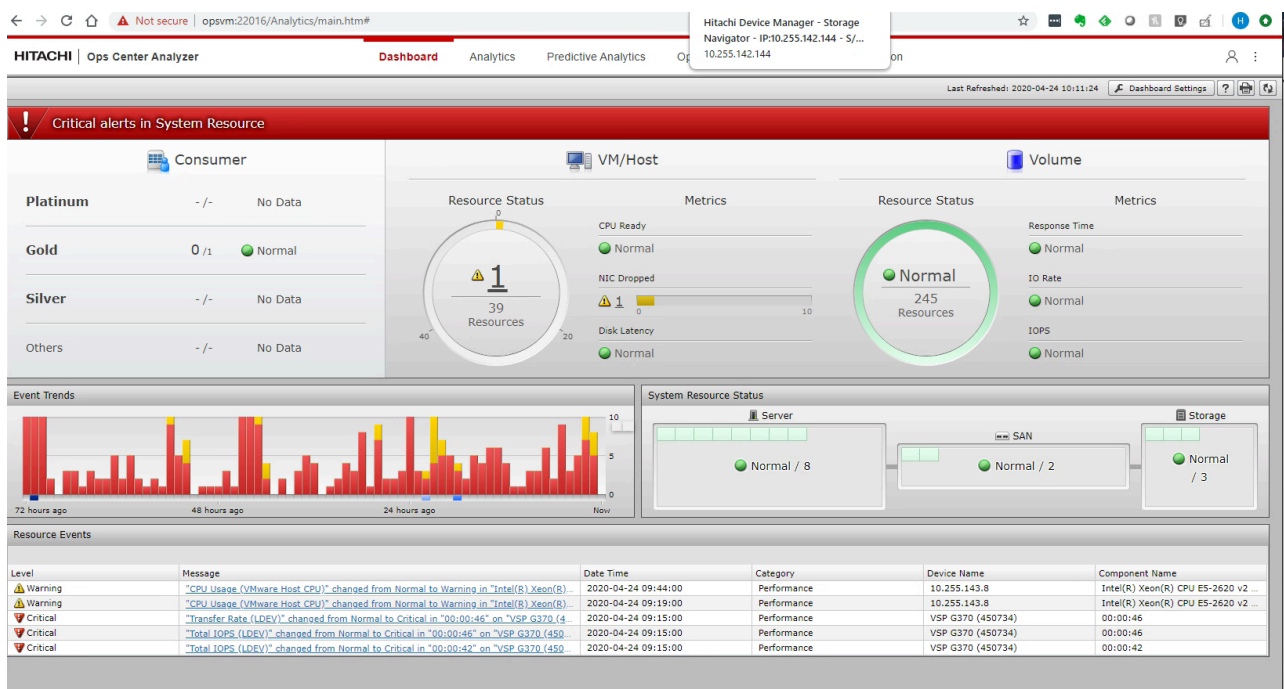
About Ops Center Analyzer

Ops Center Analyzer is a module of Ops Center that will allow one to analyze the infrastructure end to end including hosts, VM's, storage systems, and network switches. It is used to troubleshoot, predict, and find root causes for poor application performance, and resource planning.

Launching Analyzer

From the Ops Center portal, one can click on the Analyzer link to launch the Analyzer.

The Analyzer dashboard will be displayed.



From here, one can quickly see the overall health of the data center.

When there are alerts associated with managed resources, one can easily see them, and then drill down to investigate the alerts.

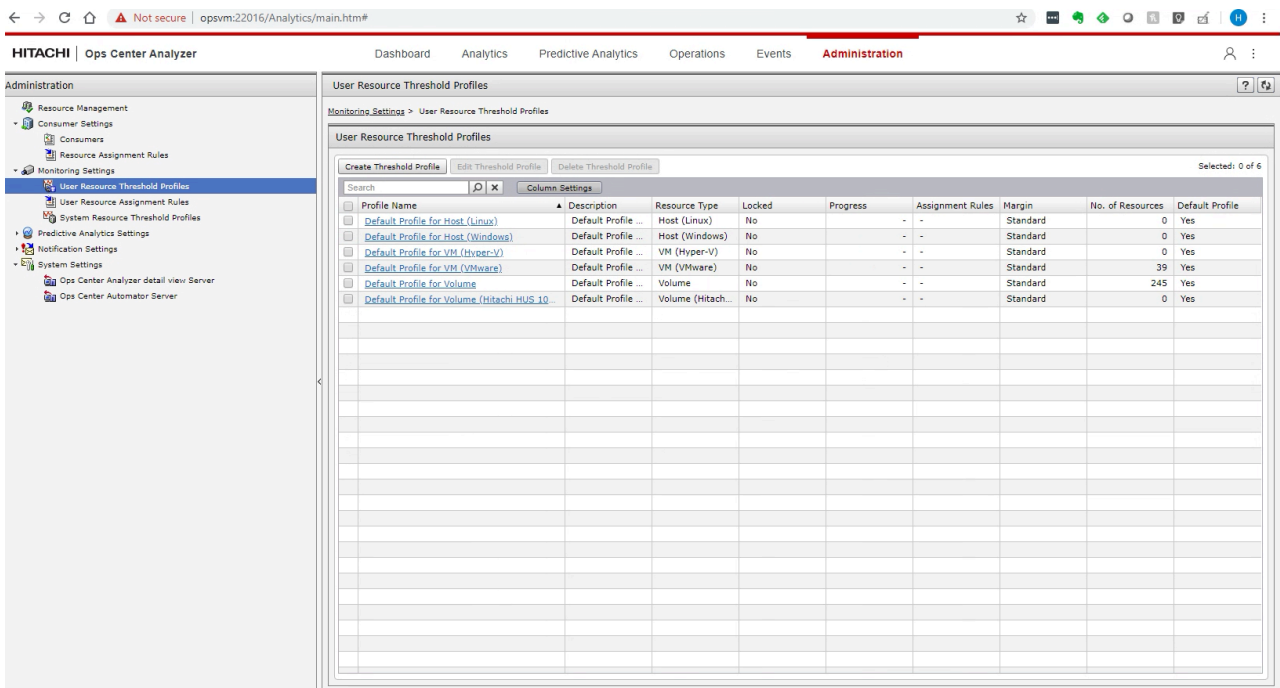
The Analyzer has built-in thresholds associated with different metrics of the resources. One can create their own thresholds. The thresholds can be either static or dynamic. The dynamic thresholds are modified over time after establishing a baseline of different metrics. For the static thresholds, one can set them manually.

When the metric of a resource exceeds the threshold, the alert is triggered.

Analyzer Administration

One can see the system defaults by going to Administration -> Monitoring Settings -> User Resource Threshold Profiles.

One should see the following display:



The screenshot shows the HITACHI Ops Center Analyzer interface. The left sidebar contains a navigation menu with categories like Resource Management, Consumer Settings, Resource Assignment Rules, Monitoring Settings, User Resource Assignment Rules, System Resource Threshold Profiles, Predictive Analytics Settings, Notification Settings, System Settings, and Ops Center Analyzer detail view/Server. The main content area is titled 'User Resource Threshold Profiles' and includes a breadcrumb 'Monitoring Settings > User Resource Threshold Profiles'. Below this, there are tabs for 'Create Threshold Profile', 'Edit Threshold Profile', and 'Delete Threshold Profile'. A table displays the existing profiles with columns for Profile Name, Description, Resource Type, Locked, Progress, Assignment Rules, Margin, No. of Resources, and Default Profile. The table lists six default profiles for Host (Linux), Host (Windows), VM (Hyper-V), VM (VMware), Volume, and Volume (Hitachi HUS 10...).

Profile Name	Description	Resource Type	Locked	Progress	Assignment Rules	Margin	No. of Resources	Default Profile
Default Profile for Host (Linux)	Default Profile ...	Host (Linux)	No	-	-	Standard	0	Yes
Default Profile for Host (Windows)	Default Profile ...	Host (Windows)	No	-	-	Standard	0	Yes
Default Profile for VM (Hyper-V)	Default Profile ...	VM (Hyper-V)	No	-	-	Standard	0	Yes
Default Profile for VM (VMware)	Default Profile ...	VM (VMware)	No	-	-	Standard	39	Yes
Default Profile for Volume	Default Profile ...	Volume	No	-	-	Standard	245	Yes
Default Profile for Volume (Hitachi HUS 10...)	Default Profile ...	Volume (Hitach...	No	-	-	Standard	0	Yes

From here, one can take a look at one of the default threshold profile.

Drilling down

There are couple different ways to trigger to End to End View.

Go to the Dashboard by clicking on the Dashboard tab on the top.

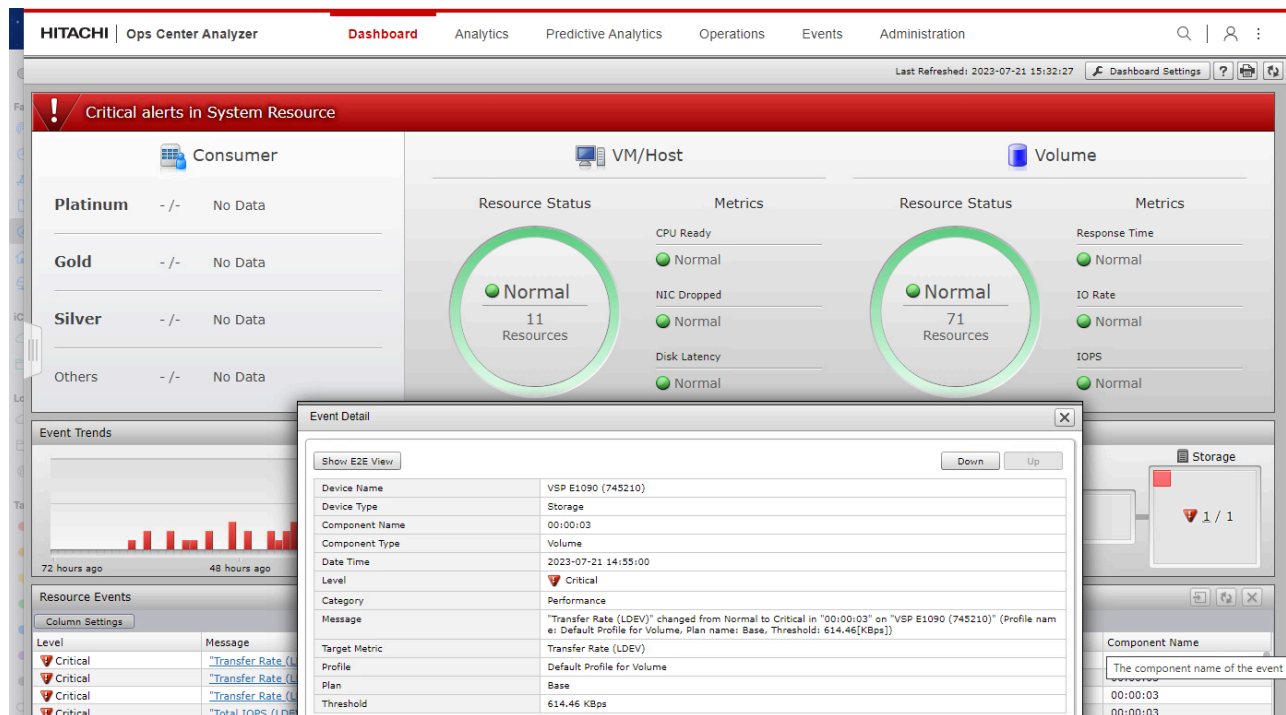
One can click on the number of the alerts in the Consumer, VM/Host, or Volume section.

Another way is to click on the message hyperlink in the Resource Events window. (All of the Events can also be seen under the Events tab)

Let's click on one of the message hyperlink in the Resource Events window. (The Resource Events pane is at the bottom right)

The Event Detail window will display.

Event Details

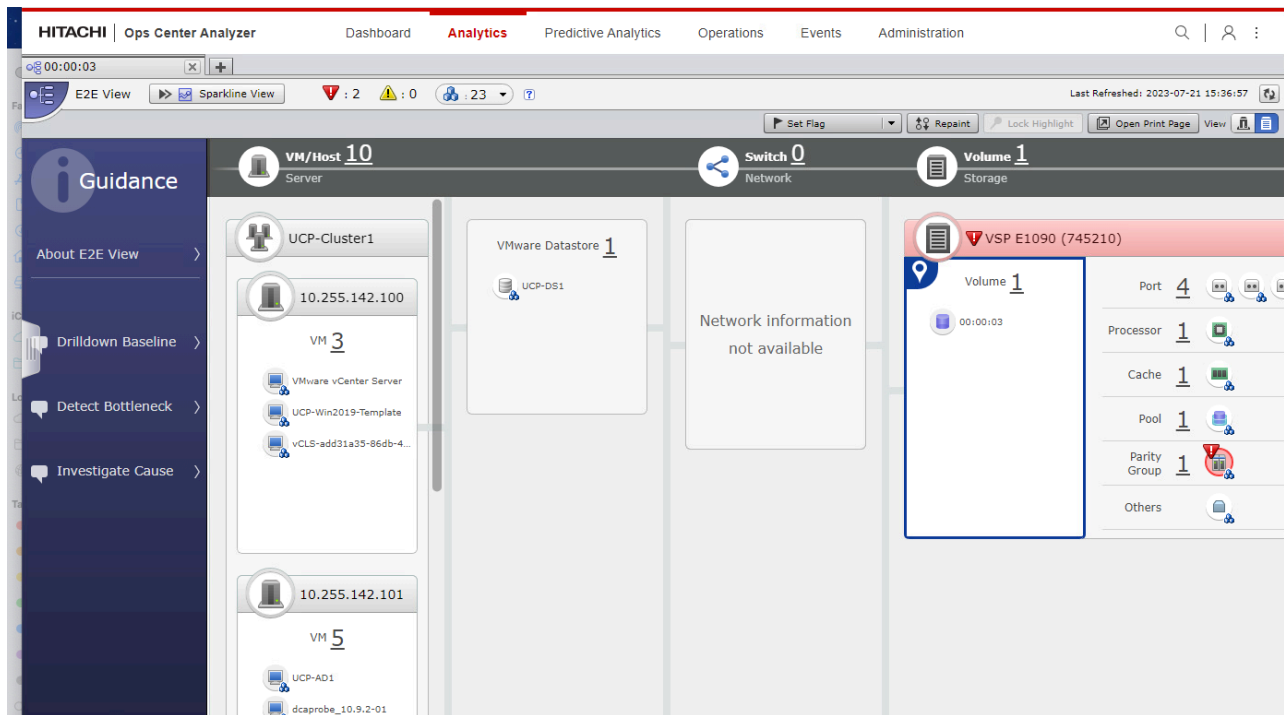


One can see the detail about the event such as the Time, Level, profile, and Threshold.

There also is a button labeled, "Show E2E View" at the top left of the window.

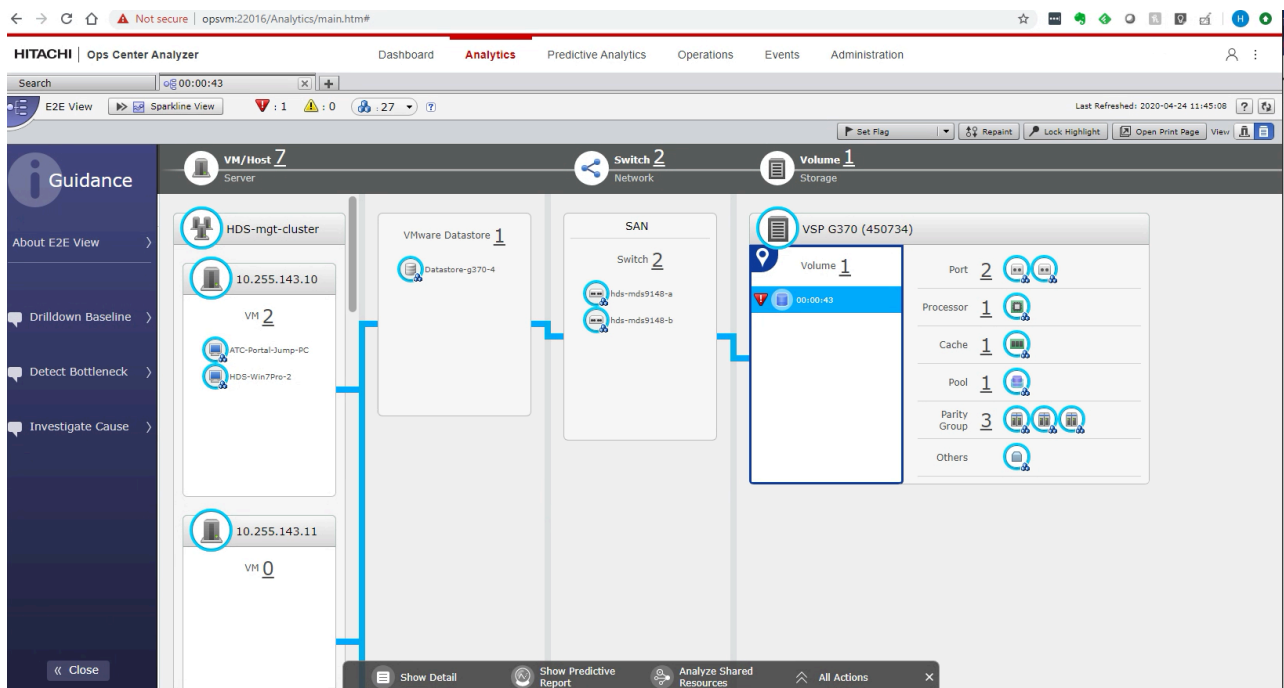
Click on that button, and the End to End View is displayed.

End to End View



From this view, one can analyze the cause of the event. Especially for the shared resource such as a storage volume.

When one clicks on the volume, it will map the other resources that are utilizing the volume.



It will also bring up a subpanel at the bottom containing "Show Detail", "Show Predictive Report", "Analyze Shared Resources", and "All Actions".

Click on "Show Detail"

It should bring up the details window.

The screenshot shows the HITACHI Ops Center Analyzer interface. A modal window titled '00:00:43' is open, displaying the 'Basic Information' tab. The window contains two sections: 'Volume Information' and 'Pool Information'.

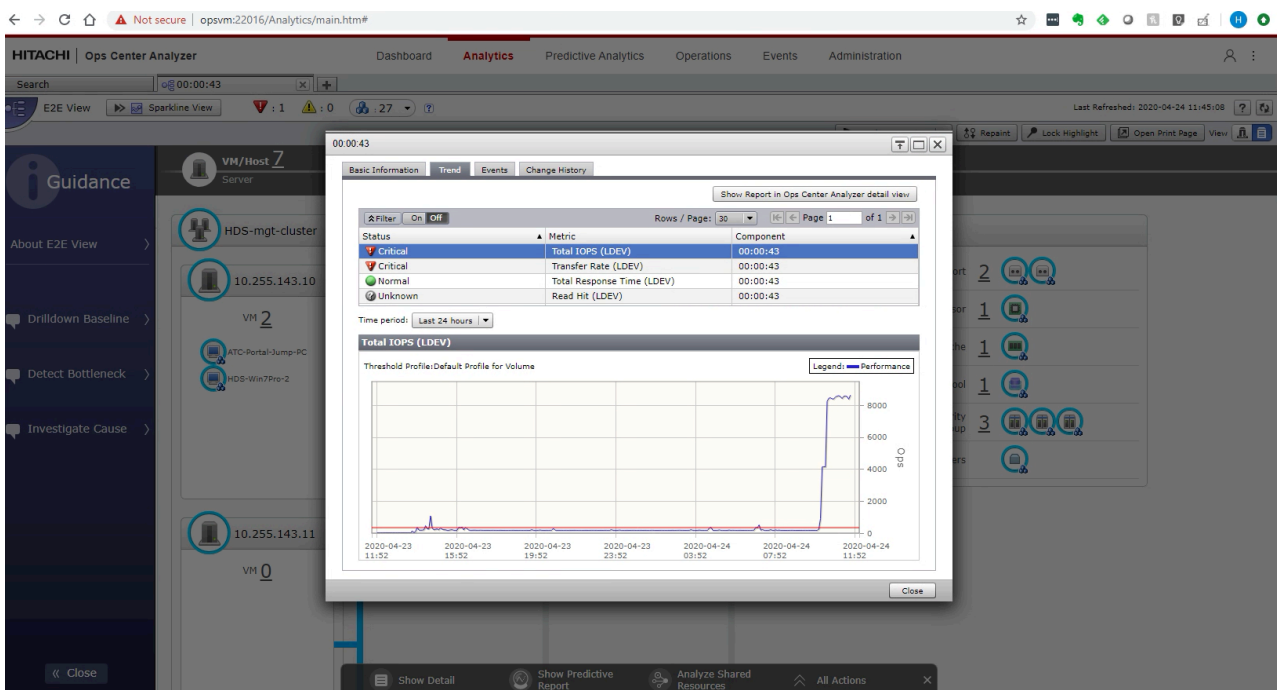
Volume Information	
Name	00:00:43
Status	Critical
Capacity	4.00 TB
Storage System	VSP G370 (450734)
Host Group	exiservers, exiservers_001, exiservers_001, exiservers_002, exiservers_002, exiservers_003, exiservers_003, exiservers_004, exiservers_004, exiservers_005, exiservers_005, exiservers_006, exiservers_006, exiservers_007, exiservers_007
Volume Location	Internal
Consumer Name	labgold
Grade	Gold
User Resource Threshold Profile	Default Profile for Volume

Pool Information	
Name	1
Status	Unknown
Capacity	29.35 TB
Capacity Used	377.22 GB
Capacity Used %	1 %
Physical Capacity	30057 GB
Physical Capacity Used	377 GB
Physical Capacity Used %	1 %
Total Efficiency Ratio	59.03 : 1
Data Reduction Ratio	1.00 : 1

Here one can see the detail of the resources.

There are other tabs in the window.

The Trend tab shows the different statistics associated with the resource



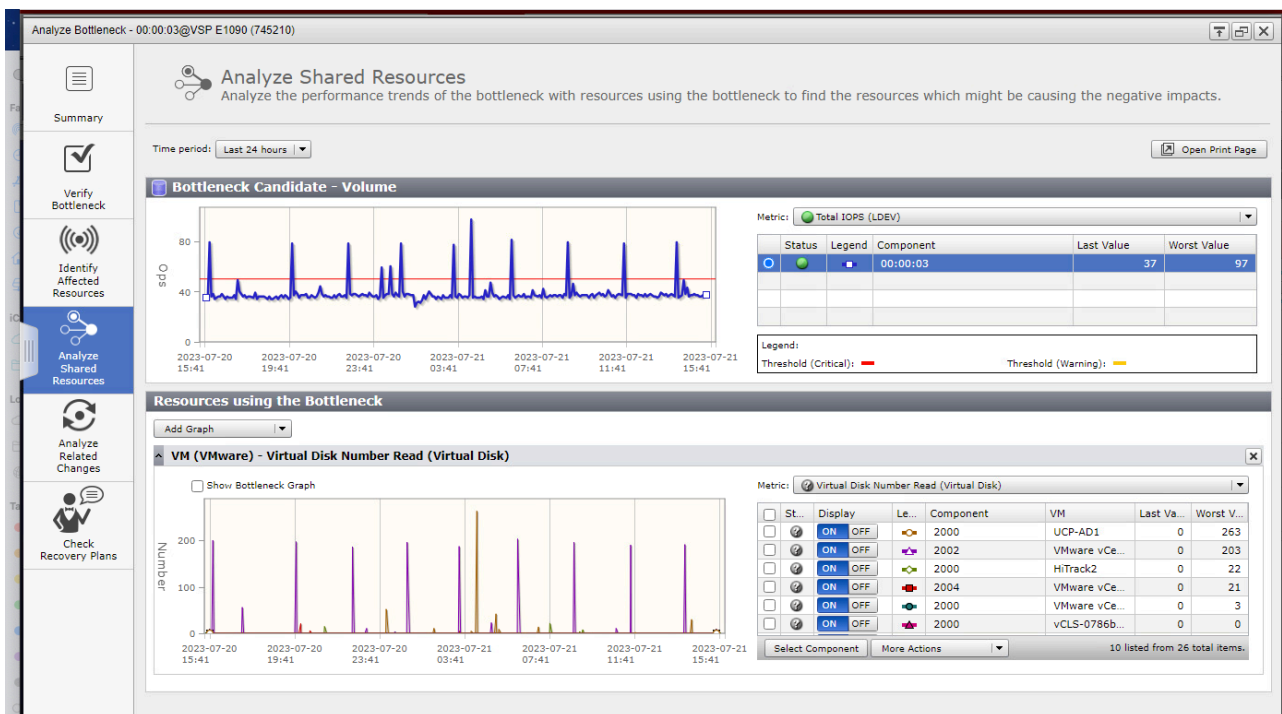
One can see easily if there is a anomaly.

The Events tab shows all the events associated with the resource



Click on the "Analyze Shared Resources"

The following window will be displayed.



One can see that the read option was causing the work load. When one selects different components on the list at the bottom right quadrant, one can find the component that was causing this alert. One can easily narrow down to the VM that was causing the alert.

The Analyzer is a powerful tool that has lot more capabilities.